1 Course Roadmap

1.1 Design Methodology Root cause

Failure → Failure analysis → Root cause analysis

Root cause can be traced from these processes:

- Design
- Material
- Manufacturing/Assembly
- Service

1.2 5 main failure types

1. distortion
   - yielding -Stress state
     -Yield criteria (Von mises, Tresca, Rankine)
   - Buckling -$F_{buckling} = \frac{n\pi^2 EI}{L^2}$
   - Creep
   - Residual stress/relaxation(not covered much in the course)
2. fracture
   - Fractography
     -brittle(intragranular - "rock-candy", intergranular - "voids")
     -ductile
     -fatigue(3 regions: crack initiation, crack propagation(beach marks, striations), brittle fracture)
   - Stress concentration factor:
     $k_I = \frac{\sigma_{max}}{\sigma_{app}} = 2\sqrt{\frac{a}{\rho}}$
   - Griffith Criterion
     $\gamma_s \equiv$ Strain energy
     $\sigma_c = \sqrt{\frac{2\gamma_s d}{\pi a}}$
     Compare $k = f\sigma\sqrt{\pi a}$ to K_{IC}
   - fatigue
     - σ_{mean}
     - S-N curve (Stress life)
     - 3 Stages (What are these?)
     - effect of dm?
       Goodman, Soderberg, Gerber
     - Paris law
3. corrosion

- EMF and Galvanic Series
  -Anodic(gives away electron) vs Cathodic (takes electron)
  What makes it more anodic?:
  cold work makes it more anodic?
  lower oxygen level?
- General corrosion
- localized corrosion:
  - Pitting
  - Crevice?
  - Grain boundary
  - Selective leeching
  - Stress corrosion cracking
    Review $\frac{da}{dr}$ vs k plot
- velocity affected
  - erosion
  - impingment?
  - cavitation
- Prevention
  - databases
  - testing
  - design
  - Cathodic - Using another material to prevent corrosion
  - anodic - Using another material to prevent corrosion

4. Wear Types of Wear:

- Abrasive -Hardness difference between two metals in contact causes plowing/cutting/fracture
  - 2 body
  - 3 body
    wear rate can be affected by:
    -particle size H_a/H_s
    -microstructure
    environment
- Adhesive
- Fretting
- Lubrication
  temperature regulation
  increases gap between surface
  wear surface can be studied using SEM, MEM, etc.

5. High T(can be included)

- creep (Study different stages and the creep rate ϵ vs t plot) insert table from homework on GB climb etc

  - Larson Miller Parameter
  - Failure mechanisms
    -grain boundary void diffusion
    -if climb is dominant, more plastic deformation?

- Thermal fatigue
- Thermal mechanical fatigue
  -Think of beam in between two walls
- Thermal shock
  $R \propto \frac{\sigma_f k}{\alpha E}$
  The smaller the thermal expansion the more resistant the material is to thermal shock

1.3 Other

1. NDT remember physical aspect on how these methods reveal failure (e.g capillary action) know pros and cons
2. Manufacturing

Focus on welding on the online lecture
There will be a question that relates last question on midterm to fatigue